
INTERESTS

Deep learning, Evolution, High Performance Computing, Cellular Automata, Generative Art

Researcher in Complex Systems and Data Science, developing adaptive AI systems that continuously learn across domains without catastrophic forgetting. With a computer science foundation and expertise in deep continual meta-learning, I bridge theoretical advances with high-performance implementations using PyTorch, JAX, and CUDA C++. My interdisciplinary approach combines insights from evolutionary computation and hardware-aware optimization to tackle complex technical challenges in designing artificially intelligent systems. <https://lokaloptima.dev/>

EDUCATION

- 2019 - 2025 **PhD in COMPLEX SYSTEMS AND DATA SCIENCE**
Vermont Complex Systems Institute, U.S.
Thesis: “Zap it, Tune it, Learn it: Neural Resampling for Efficient Meta-Learning in Transfer, Few-Shot and Continual Learning”
Supervisor: Nick CHENEY
- 2015 - 2018 **Master Degree in COMPUTER SCIENCE**
University of Pisa, Italy
110/110 *summa cum laude*
Thesis: “Vision-based Deep Learning Model for Guiding Multi-fingered Robotic Grasping”
Advisor: Davide BACCIU
- 2016 - 2017 **Erasmus Programme**
University of Southampton, UK
First Class Honours
- 2012 - 2015 **Bachelor Degree in COMPUTER SCIENCE**
University of Pisa, Italy
110/110 *summa cum laude*
Thesis: “Prediction of Protein Solvent Accessibility using Neural Networks”
Advisors: Prof. Alessio MICHELI and Prof. Gianluca POLLASTRI
- 2014 - 2015 **Erasmus Programme**
University College Dublin, Ireland
First Class Honours

PUBLICATIONS

- Frati, L., Traft, N., Clune, J., and Cheney, N. (2025). How Weight Resampling and Optimizers Shape the Dynamics of Continual Learning and Forgetting in Neural Networks. *arXiv preprint arXiv:2507.01559*.
- Petak, C., Frati, L., Vrooman, R. M., Pespeni, M. H., and Cheney, N. (2025). The variability of evolvability: properties of dynamic fitness landscapes determine how phenotypic variability evolves. In *Proceedings of the National Academy of Sciences* 122.51 (2025)
- Frati, L., Traft, N., Clune, J., and Cheney, N. (2024). Reset it and forget it: Relearning last-layer weights improves continual and transfer learning. In *ECAI 2024* (pp. 2998-3005). IOS Press.
- Frati, L., Petak, C., and Cheney, N. (2024). Networks of Binary Necklaces Induced by Elementary Cellular Automata Rules. In *Artificial Life Conference Proceedings* 36 (Vol. 2024, No. 1, p. 117). MIT Press.
- Frati, L., Traft, N., and Cheney, N. (2023). Omnimage: Evolving 1k image cliques for few-shot learning. In *Proceedings of the genetic and evolutionary computation conference* (pp. 476-484).
- Petak, C., Frati, L., Pespeni, M. H., and Cheney, N. (2023). Coping with seasons: evolutionary dynamics of gene networks in a changing environment. In *Proceedings of the Companion Conference on Genetic and Evolutionary Computation* (pp. 163-166).

- Petak, C., Frati, L., Brennan, R. S., and Pespeni, M. H. (2023). Whole-genome sequencing reveals that regulatory and low pleiotropy variants underlie local adaptation to environmental variability in purple sea urchins. *The American Naturalist*, 202(4), 571-586.
- Powers, J., Grindle, R., Frati, L., and Bongard, J. (2021). A good body is all you need: Avoiding catastrophic interference via agent architecture search. In *35th Conference on Neural Information Processing Systems (NeurIPS 2021)* arXiv:2108.08398.
- Beaulieu, S., Frati, L., Miconi, T., Lehman, J., Stanley, K. O., Clune, J., and Cheney, N. (2020). Learning to continually learn. In *ECAI 2020* (pp. 992-1001). IOS Press.
- Powers, J., Grindle, R., Kriegman, S., Frati, L., Cheney, N., and Bongard, J. (2020). Morphology dictates learnability in neural controllers. In *Artificial Life Conference Proceedings 32* (pp. 52-59). MIT Press.

EXPERIENCE

- 2026 - 2027 **Research Engineer** Minerva Project, Tübingen AI Center
- 2025 **AI Consultant** Genomate Health
- 2024 - 2025 **AI Tooling & Game Developer** Downbeat Interactive
- 2020 - 2025 **Reviewer:** ALife, GECCO, CoLLAs, ECAI
- 2023 **Artist** for the “Co-Create: The Artist in the Age of Intelligent Machines” exhibition @ BCA
- 2020 - 2022 **Instructor** and founder of *P5.js club* @ UVM
- 2021 **Advisor** for the *Art + AI group* led by Jen Karson @ UVM

SERVICE

- 2020 - 2021 **Committee member** for master student Jeechieu Ta @ Drexel University
- 2022 **Invited speaker** for the “Liquid Architectures + Leaky Territories Workshops” @ UVM
- 2022 **Invited speaker** for *Modeling Complex Systems* class by Laurent Hébert-Dufresne @ CSDS
- 2022 **Invited speaker** for the *Society of Women in Computer Science (SWICS)* @ UVM
- 2021 **Advisor** for the *Art + AI group* led by Jen Karson @ UVM
- 2015 **Mentor** for Pisa CoderDojo - student organized programming workshops for young people
- 2012 **Mentor** for Students with Disabilities @ University of Pisa

CONFERENCES & WORKSHOPS

- 2025 **ALICE** Copenhagen, poster: The Challenges of Life, project: Neuromorphic Cellular Automata
- 2025 **AI Summit 2025** Budapest, Hungary
- 2024 **ECAI** European Conference on Artificial Intelligence
- 2024 **aiVermont**, Education + AI Summit
- 2024 **ALife** International Conference for Artificial Life (ALife)
- 2023 **Creative AI Vermont** Symposium on Artificial Intelligence + Art
- 2023 **Gecco** Genetic and Evolutionary Computation Conference
- 2022 - 2024 **SRC** Student Research Conference @ UVM
- 2020 **ECAI** European Conference on Artificial Intelligence
- 2020 **ALife** International Conference for Artificial Life (ALife)
- 2019 **DARPA L2M** Lifelong Learning Machines meeting

SKILLS

- DL** PyTorch, Jax, **CUDA C++ certificate**, MLX, Numba
- Tools** linux, neovim, bash/zsh, ssh, SLURM, conda, uv, git, LaTeX
- Languages** ITALIAN: Native ENGLISH: C2 FRENCH: A2 HUNGARIAN: A1